The mission's primary scientific objective is to investigate Titan's prebiotic chemistry – the complex organic molecules that may offer clues about how life can arise in environments beyond Earth. Dragonfly carries instruments for mass spectrometry, seismic sensing, meteorology, gamma-ray neutron spectroscopy, and high-resolution imagery. These tools will allow the spacecraft to analyze surface composition, atmospheric structure, geological history, and potential habitability. By studying Titan's organics and subsurface ice/ocean interface, Dragonfly aims to answer key questions about life's chemical origins in conditions different from Earth's.

Dragonfly is also designed for high autonomy, given the long communication delays between Titan and Earth. It must plan routes, avoid hazards, monitor power reserves, and adapt to weather changes independently. The mission will be powered by a radioisotope thermo-electric generator (RTG), ensuring energy availability during Titan's long nights and low sunlight conditions.

Figure 9.12 shows the NASA Dragonfly rotorcraft designed for Titan surface exploration. The top view highlights the octocopter configuration, with eight rotors arranged symmetrically to provide stability and lift in Titan's dense atmosphere. The side view illustrates the aerodynamic fuselage, landing skids for uneven terrain, and instrument housing for environmental sampling and autonomous navigation. The dual-rotor arm layout ensures redundancy, fault tolerance, and maneuverability during repeated takeoff and landing sequences across diverse surface regions on Titan.